# Page Checks — Technical Reference

A detailed, engineering-oriented breakdown of every page-level check in the Sitecore AI Page Search Readiness Checker: what it inspects, the exact thresholds and scoring math, and where it lives in the codebase.

**Source of truth:** `src/lib/rule-engine/engine.ts` (check definitions), `helpers.ts` (shared utilities), `types.ts` (interfaces), `site-score.ts` (site-level aggregation).

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## Scoring model

- Each check returns a status of `pass` / `warning` / `fail` and a numeric score `0..max`.
- Page total = sum of enabled check scores = **81 points** across three pillars:

- **SEO**: 35 pts (6 scored + 1 info-only)

- **AEO**: 23 pts (4 scored)

- **Content Quality (GEO)**: 23 pts (5 scored)

- Info-only checks contribute `0` to the total but emit findings/guidance.
- A **gate check** runs before scoring and short-circuits the whole panel if the page is intentionally excluded from indexing.

### SEO / AEO / GEO definitions (technical framing)

| Pillar | Optimizes for | Primary consumers | Signal type |
|---|---|---|---|
| **SEO** | Crawl + index + SERP rendering | Googlebot, Bingbot | HTML `<head>` metadata, document structure, link graph |
| **AEO** | Featured snippets, voice, answer boxes | Google answer surfaces, voice assistants | Structured data (JSON-LD), Q&A formatting |
| **GEO** | LLM retrieval & citation | ChatGPT, Gemini, Claude, Perplexity, AI Overviews | Content originality, readability, semantic HTML, performance |

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## Gate check — noindex / nosnippet

**Location:** `engine.ts:32–71`

**Logic:** Before any scoring, evaluate exclusion signals:

1. Sitecore `noindex` field set on the item, **OR**

2. `<meta name="robots">` / `<meta name="googlebot">` containing `noindex`, `nosnippet`, or `max-snippet:0`.

**Behavior:** If triggered, the panel is blocked and surfaces a single message; no checks are scored. Rationale: optimizing a non-indexable page produces misleading scores.

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## SEO checks (35 pts)

### 1. `seo-meta-title` — Meta Title Length

- **Location:** `engine.ts:78`
- **Inspects:** Character length of the page title field.
- **Scoring (max 8):**

- `pass` 8 — length ∈ [50, 60]

- `warning` 5 — length ∈ [30, 49] ∪ [61, 70]

- `fail` 2 — length < 30 or > 70

- **Rationale:** ~600px SERP truncation budget; titles outside the band are truncated or under-utilized.

### 2. `seo-meta-description` — Meta Description Present + Length

- **Location:** `engine.ts:154`
- **Inspects:** Presence and length of the meta description.
- **Scoring (max 8):**

- `pass` 8 — length ∈ [120, 160]

- `warning` 4 — present but outside range

- `fail` 0 — missing

- **Rationale:** Controls SERP snippet copy; absent → Google auto-generates from body.

### 3. `seo-headings` — H1 Count + Heading Hierarchy

- **Location:** `engine.ts:212`
- **Inspects:** Count of `<h1>` and whether heading levels descend without skips.
- **Scoring (max 6 = 3 + 3):**

- H1 sub-score (3): exactly one H1 → 3; zero or 2+ → 0–1

- Hierarchy sub-score (3): no skipped levels (e.g., H2→H4) → 3; skips present → 0

- **Rationale:** Single H1 = unambiguous topic; ordered hierarchy = parseable document outline for crawlers and assistive tech.

### 4. `seo-image-alt` — Image Alt Text Coverage

- **Location:** `engine.ts:288`
- **Inspects:** Ratio of `<img>` elements with non-empty `alt`.
- **Scoring (max 5):** `round((imagesWithAlt / totalImages) * 5)`

- `pass` 5 — 100% coverage

- `warning` 1–4 — partial

- `fail` 0 — none (or no images → treated as full)

- **Rationale:** Image search indexing + WCAG accessibility.

### 5. `seo-canonical` — Canonical Tag

- **Location:** `engine.ts:365`
- **Inspects:** Presence and target of `<link rel="canonical">`.
- **Scoring (max 5):**

- `pass` 5 — present and self-referencing

- `warning` 2 — present but points to a different URL

- `fail` 0 — missing

- **Rationale:** Duplicate-content consolidation; prevents ranking-signal dilution across URL variants.

### 6. `seo-og-tags` — Open Graph Tags

- **Location:** `engine.ts:426`
- **Inspects:** Presence of `og:title`, `og:description`, `og:image`.

- **Scoring (max 3):** score = count of present tags

- `pass` 3 — all three; `warning` 2 — two of three; `fail` 0–1 — fewer

- **Rationale:** Controls social/link-unfurl preview cards (LinkedIn, Slack, Facebook).

### 7. `seo-anchor-text` — Internal Link Anchor Text Quality *(info-only, 0 pts)*

- **Location:** `engine.ts:472`
- **Inspects:** Internal anchors against a generic-phrase blocklist ("click here", "read more", "learn more", etc.; see `helpers.ts`).
- **Output:** Count and list of generic anchors; no score.
- **Rationale:** Anchor text is a relevance signal; generic anchors waste it.

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## AEO checks (23 pts)

### 8. `aeo-faq-schema` — FAQPage JSON-LD

- **Location:** `engine.ts:511`
- **Inspects:** FAQ accordion items in content; **generates** a `FAQPage` JSON-LD block ready to paste.
- **Scoring (max 8):** full score when no FAQ items present, otherwise emits generated schema.
- **Rationale:** Eligibility for FAQ rich results and direct answer extraction.

### 9. `aeo-faq-answer-quality` — FAQ Answer Length & Quality

- **Location:** `engine.ts:553`
- **Inspects:** Each FAQ answer for: plain text (no HTML), length ∈ [40, 300] chars, self-contained.
- **Scoring (max 5):** `(qualifyingAnswers / totalAnswers) * 5`
- **Rationale:** Snippet/voice extraction favors concise, standalone answers.

### 10. `aeo-organization-schema` — Organization JSON-LD

- **Location:** `engine.ts:702`
- **Inspects:** Generates a ready-to-paste `Organization` JSON-LD snippet (name, logo, etc.).
- **Scoring (max 5):** always full — provides the snippet for the global template.
- **Rationale:** Knowledge-panel eligibility and entity disambiguation.

### 11. `aeo-structured-data` — Rich Results Eligibility (Breadcrumbs)

- **Location:** `engine.ts:728`
- **Inspects:** Derives `BreadcrumbList` JSON-LD from the URL path.
- **Scoring (max 5):** always full — provides generated snippet.
- **Rationale:** Breadcrumb SERP rendering + site-hierarchy signal.

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## Content Quality / GEO checks (23 pts)

### 12. `cq-uniqueness` — Unique Content Signal

- **Location:** `engine.ts:787`
- **Inspects:** LLM (Claude) classification of body content as `unique` / `borderline` / `commodity`.
- **Scoring (max 10):** `unique` → 10 ( `pass` ), `borderline` → 5 ( `warning` ), `commodity` → 0 ( `fail` ).
- **Rationale:** Originality is the dominant GEO/E-E-A-T signal; highest single weight by design.

### 13. `cq-page-experience` — Performance Score

- **Location:** `engine.ts:966`
- **Inspects:** PageSpeed Insights score + Core Web Vitals (external API).
- **Scoring (max 5):**

- `pass` 5 — PSI ≥ 70

- `warning` 3 — PSI ∈ [50, 69]

- `fail` 0 — PSI < 50 or page not crawlable

- **Rationale:** Confirmed ranking factor; UX proxy.

### 14. `cq-agentic-readiness` — Semantic Landmarks

- **Location:** `engine.ts:1048`
- **Inspects:** Presence of HTML5 landmarks `<main>`, `<article>`, `<nav>`, `<header>`; also flags interactive elements missing `aria-label`.
- **Scoring (max 4):** score = count of landmarks present (0–4)

- `pass` 4 — all four; `warning` 3; `fail` 0–2

- **Rationale:** Agent/assistant DOM navigation; isolates primary content from chrome.

### 15. `cq-reading-grade` — Passage-level Reading Grade

- **Location:** `engine.ts:1118`
- **Inspects:** Automated Readability Index (ARI) per paragraph (paragraphs ≥ 20 words).

- `ARI = 4.71*(chars/words) + 0.5*(words/sentences) - 21.43`

- **Scoring (max 4):** `(paragraphsAtGrade≤13 / totalParagraphs) * 4`

- `pass` 4 — all ≤ grade 13; `warning` 3 — ≥75%; `fail` 0–2 — <75%

- **Rationale:** Readability correlates with dwell time and clean LLM summarization/quotation.

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## Disabled checks (defined, commented out in the orchestrator)

These exist in code but are not currently part of the 81-pt total. Useful to know for roadmap discussions.

| Check ID | Name | Max | Location | Notes |
|---|---|---|---|---|
| `aeo-howto-schema` | HowTo Schema | 7 | `engine.ts:626-698` | Validates HowTo JSON-LD step `name` / `text` |
| `aeo-question-headings` | Question-format H2s | 0 (info) | `engine.ts:752` | Regex count of interrogative H2s |
| `cq-author` | Author Present + Linked | 6 | `engine.ts:845` | linked → 6 / name only → 3 / missing → 0 |
| `cq-freshness` | Published Date + Freshness | 5 | `engine.ts:897` | ≤365d → 5 / ≤730d → 2 / >730d → 0 |
| `cq-followup-gaps` | Content follow-up gaps | 0 (info) | `engine.ts:1193-1228` | LLM-generated follow-up questions vs. coverage |

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## Site-level scoring (aggregate, not page-level)

**Location:** `site-score.ts:307-366`. Rolls page-derived HTML signals into 5 site pillars. Listed for completeness — these are separate from the per-page panel.

| Pillar | Max | Inputs |
|---|---|---|
| Crawlability | 15 | AI-crawler permissions, canonical, robots meta |
| Structured | 0–20 | Conditional schema presence (FAQPage, Article, BreadcrumbList, Organization) |
| Answerability | 20 | H1 count, question H2s, reading grade, paragraph length |
| Authority | 25 | AI uniqueness, content depth, freshness, anchor-text quality |
| Retrievability | 20 | Title length, meta description, OG tags, landmarks, image alt coverage |

## Key files

| File | Responsibility |
|---|---|
| `src/lib/rule-engine/engine.ts` | All check definitions, scoring logic, gate (lines 32–1324) |
| `src/lib/rule-engine/helpers.ts` | ARI, word/char counts, anchor blocklist, shared parsing utils |
| `src/lib/rule-engine/types.ts` | Check/result interfaces and status enums |
| `src/lib/rule-engine/site-score.ts` | Site-level aggregation pillars |